

Note 1.2 Row Echelon Form

The Definition of Row Echelon Form

- **lead variables:**
variables corresponding to the first nonzero elements in each row of the reduced matrix.
- **free variables:**
variables corresponding to the skipped column in the reduction process.
- **definition**
 - The first nonzero entry in each nonzero row is 1.
 - The row k that consists of not entirely zeros must have less number of the leading zero entries than the row $k + 1$.
 - Rows whose entries are all zero are below the rows having nonzero entries.
 - Example:

$$\begin{pmatrix} 1 & 4 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

are in the row echelon form and

$$\begin{pmatrix} 2 & 4 & 6 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

are not in the row echelon form.

- **Gaussian elimination:**
The process of using [elementary row operations](#) to transform a linear system into one whose augmented matrix is in row echelon form.(reference: [Elimination](#))

Overdetermined Systems

- **Definition:** A linear systems whose equations are more than unknowns. (In matrix, suppose that there are m rows and n columns, then $m > n$.)
- **Features:** usually inconsistent.(not always)
- **Example:**

$$\left(\begin{array}{cc|c} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 2 & 1 & 3 \end{array} \right)$$

Underdetermined Systems

- **Definition:**

A linear system whose equations are less than unknowns. (In matrix, suppose that there are m rows and n columns, then $m < n$.)

- **Features**

- usually consistent and not possible to have a unique solution.
- Suppose that there are m linear equations in n unknowns and r nonzero rows, there will be r lead variables and $n - r$ free variables, where $n - r > 0$.

- **Example**

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & -1 & 1 & 2 \end{array} \right)$$

Reduced Row Echelon Form

- **Definition**

- In row echelon form.
- The first nonzero entry in each row is the only nonzero entry in its column.

- **Example**

$$\left(\begin{array}{cccc} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

Application 1 Traffic Flow

- **Strategy:** The number of entering equals to the number of leaving at each intersection.
- **Example:** [The example of traffic flow](#)

Application 2 Electrical Networks

- **Strategy:** Kirchhoff's Laws: the sum of inputting currents equals the sum of outputting currents at every node and the algebraic sum of volt gains equals to the algebraic sum of volt drops.
- **Example:** [The example of electrical networks](#)

Homogeneous Systems

- **Definition:** A system of linear equations whose constants on the right hand side are all zero.

- **Feature:** Always consistent and has a solution that is all variables equal to zero.
- **Theorem 1.2.1:** A $m \times n$ homogeneous system of linear equations has a nontrivial solution if $n > m$.
- **Remark:** A $m \times n$ homogeneous system of linear equations has a unique solution that is all variables equals to zero.

Application 3 Chemical Equations

- **Strategy:** Use homogeneous system to find the nontrivial solution to balance the chemical equation.

Application 4 Economic Models for Exchange of Goods

- **Strategy:** Turn the economic model that the inputting value equals to the outputting value into a linear system of equations.

Section 1.2 Exercises

- [question 7th](#)
 - The equations in three unknowns in a homogeneous linear system are planes in the three-dimensional space intersecting the original point.
If a equation is the multiple of another, then the two plane are indeed the same plane. Then they will have infinitely many solutions on the plane.
If a equation is not the multiple of another, then the two plane will intersect at a line crossing the original point. All the points on the line is the solutions so they will also have infinitely many solutions.
 - There might be no solutions or infinitely many solutions.
The equations in a non-homogeneous 2×3 linear system are the planes in the three-dimensional space not intersecting the original point.
The two plane might be parallel, intersect in a line or be the same plane. When parallel, they have no solutions. When intersecting in a line or being the same plane, they have infinitely many solutions.